SOE YAN NAING

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Work authorization: S Pass eligible · Serving 3-year MOE Tuition Grant bond · Available immediately

PROFILE

Information Technology graduate who builds, tests and deploys full-stack web applications and Python-based AI tools. Experienced with React, Next.js, Node.js, TypeScript, PostgreSQL and REST APIs, with a working habit of covering code with automated tests and applying secure coding practices.

EDUCATION

Temasek Polytechnic *Diploma in Information Technology | Apr 2026*
CGPA 3.39 / 4.00 · Relevant coursework: Machine Learning for Developers (A), Network and Cloud Technology (A), Data Structures and Algorithms (B+), Full Stack Web Development (B+), DevOps Essentials (B+), Application Security (B+)

Major Project: Edulift V2 (B+)

TECHNICAL SKILLS

- **Languages & Frontend:** TypeScript, JavaScript, Python, HTML/CSS, React, Next.js
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, PostgreSQL (Drizzle ORM, Neon), Pydantic
- **AI & Data:** RAG, LLM tool calling, OpenAI and Gemini APIs, Pandas, Scikit-learn
- **Cloud & DevOps:** AWS (EC2, Lambda, IAM, SES), Docker, Kubernetes (AKS), Jenkins, Grafana, Vercel, Streamlit Cloud
- **Testing & Security:** Jest, Cypress, Git/GitHub, secure coding practices, Zod, authorization controls, session authentication

EXPERIENCE

Temasek Polytechnic ITSM, Singapore
Digital Ambassador Intern | Apr 2025 – Aug 2025

- Automated departmental compliance reporting and IT asset tracking by replacing manual checks with Microsoft Power Automate approval workflows, saving time and manual workload for senior staff.
- Resolved L1/L2 software, network, and access-control incidents in ServiceNow, ~30 tickets per week, meeting SLA targets.
- Analyzed operational logs to find recurring bottlenecks, supported CI/CD deployment testing, and documented runbooks.
- Coordinated with vendors on hardware replacement and warranty claims, giving end-users and stakeholders regular updates.

PROJECTS

Full-Stack Financial Dashboard & Banking App | *React, Node.js, Express.js, PostgreSQL, Jest*

- Prevent duplicate transactions by implementing API idempotency controls, with route guards and Clerk authentication.
- Wrote multi-table PostgreSQL queries with ACID transactions, covered by automated Jest unit and integration tests.

Full-Stack Movie Recommendation Platform | *Next.js, React, JavaScript, PostgreSQL, Jest, Cypress*

- Integrated the TMDB API and tuned PostgreSQL queries and caching for faster media discovery.
- Validated components and end-to-end user journeys with Jest and Cypress test suites.
- Identified an IDOR authorization vulnerability during code review and fixed it by enforcing ownership verification and adding Zod schema validation across all server action inputs.

IT Helpdesk Chatbot & RAG Pipeline | *Python, PostgreSQL, OpenAI API*

- Built a first-line IT support bot that answers queries from a technical knowledge base and creates structured tickets through LLM tool calling when escalation is needed.
- Combined BM25 keyword search with dense vector retrieval using Reciprocal Rank Fusion and added an LLM-as-a-judge check to reduce hallucinated answers.

AI-Powered Slide Generation System (Edulift V2) | Final Year Project | *Python, Streamlit, Gemini API, python-pptx*

- Built a Streamlit app that converts unstructured text into structured PowerPoint decks under 2 minutes with relevant imagery.
- Enforced strict JSON schema validation and automated error recovery across multi-step LLM calls, reducing malformed slide output and runtime failures.

Cloud & DevOps Coursework Projects | Docker, Jenkins, Kubernetes (AKS), Grafana, AWS
Temasek Polytechnic - DevOps Essentials, IoT Application Development

- Containerised applications and deployed them to Azure Kubernetes Service through Jenkins CI/CD pipelines, with Grafana dashboards for monitoring.
- Deployed IoT backend services on AWS EC2 with Lambda event handlers, IAM roles for access control, and SES for automated email notifications.